Supplementary Material

Does an Asymmetric-Reward E-DDQN Improve Safety-Prioritized Cybersickness Detection in Virtual Reality? A Leakage-Controlled Comparison with Standard DQN and Cost-Sensitive Baselines

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S1 Data, provenance and preprocessing

S1.1 Provenance

The evaluation data are the 35 sessions of the UFFCSData curated release (`DATABASE_SELECTED_DATA`, December 2019, seated posture; 9,390 samples at ≈1 Hz). Each curated file was matched sample by sample to one raw export (`RAW_DATABASE`); `dataset.csv`, the derived file used in the earlier version of this study, is exactly these sessions concatenated in lexicographic folder order (1, 10, 11, . . .), with integer-coded categories and no session identifier. The 37 October-2019 standing-posture raw sessions were used only for code development and budget calibration; their camera-orientation channels are constant within every session (a logging defect). Two December-2019 raw sessions that the dataset authors excluded were not used. Full per-session table: `results/preprocessing/session_table.csv`.

S1.2 Label

$y_t = 1$ iff the self-reported discomfort level (0–3) at second t is ≥ 1. Prevalence 44.2 % (Flight 41.2 %, Race 48.1 %); 9 of 35 sessions never report discomfort; the median onset is 82 s after the start. Labels are close to step functions within a session (on average 1.1 binary switches per session), so consecutive samples are strongly dependent.

S1.3 Linkage groups

No participant identifier exists. Sessions are linked when their nine-field questionnaire profile is identical or when one restarts within 180 s after an aborted session (< 200 samples) with at most two differing fields; this gives 21 groups (log: `results/preprocessing/linkage_log.csv`). Groups are the unit of every split.

S1.4 Causal features

Definitions of all 39 telemetry features and 11 questionnaire indicators: `results/preprocessing/feature_dictionary.csv`. The models use preprocessing (imputation value, spike threshold, scaler centre and scale) fitted on the training rows of each inner split. For documentation, `learned_parameters_by_outer_fold.csv` lists the imputation value and the spike threshold fitted on each outer fold's development groups, together with the fixed 80-fps frame-drop threshold (the scaler parameters are not listed); no test rows enter either fit. Executable checks: truncation invariance of all features (max. difference 0.0 over 40 random cut points); bit-identical preprocessing parameters when test rows are tampered with; zero session, group, identical-profile and duplicated-feature-row overlap between every development set and its test fold (`partition_overlap_checks.csv`).

S1.5 Removed variables

Field of view, static-frame and auto-camera flags (constant within each game: duplicates of the game indicator); haptic feedback, degree of control, DOF simulation, locomotion, posture and symptoms (constant overall); absolute player position (encodes course distance and therefore elapsed time; absent in the October wave); VRSQ scores (post-session).

S2 Protocol, search spaces and selected configurations

The protocol (`configs/protocol.json`) and the perturbation configuration (`configs/attacks.json`) were committed to the repository before any model was trained on the evaluation sessions (commit recorded in `configs/FROZEN_PROTOCOL_COMMIT.txt`). Outer-test rows are not even featurised during hyper-parameter search; the final stage scores each outer-test fold once.

S2.1 Search spaces

Family	Hyper-parameter	Candidate values
All neural models	feature_set	telemetry, telemetry+profile, exposure, exposure+profile
All neural models	hidden	[64, 64], [128, 128], [256, 256], [256, 512, 256]
All neural models	dropout	0.0, 0.1, 0.2, 0.3
All neural models	norm	batch, layer
All neural models	lr	0.0003, 0.001, 0.003
All neural models	weight_decay	0.0, 0.0001, 0.001
All neural models	batch_size	256, 512
All agents	eps_decay	0.9, 0.95, 0.97
All agents	warmup	0, 4000
All agents	updates_per_learn	1, 2, 4
E-DDQN	gamma	0.0, 0.5, 0.9, 0.99
E-DDQN	tau	0.005, 0.01, 0.05
E-DDQN	rho	0.4, 0.5, 0.6
E-DDQN	full_info	False, True
E-CB	rho	0.4, 0.5, 0.6
E-CB	full_info	False, True
DQN	gamma	0.0, 0.5, 0.9, 0.99
DQN	hard_every	10, 100, 500
CS-MLP	sampling	uniform, balanced
LogReg	feature_set	telemetry, telemetry+profile, exposure, exposure+profile
LogReg	C	0.0001, 0.001, 0.01, 0.1, 1.0, 10.0
HGB	feature_set	telemetry, telemetry+profile, exposure, exposure+profile
HGB	learning_rate	0.02, 0.05, 0.1
HGB	max_iter	50, 100, 200, 400
HGB	max_leaf_nodes	4, 8, 16, 31
HGB	min_samples_leaf	20, 100, 300
HGB	l2_regularization	0.0, 1.0, 10.0
RF	feature_set	telemetry, telemetry+profile, exposure, exposure+profile
RF	max_depth	3, 5, 8, None
RF	min_samples_leaf	5, 25, 100, 300
RF	max_features	sqrt, 0.5, 1.0
RF	n_estimators	300
All agents	fixed	epochs=40, eval_every=2, chunk=256, learn_every=2, consolidation=5, eps_start=1.0, eps_end=0.01, buffer=50000, grad_clip=1.0

Number of distinct configurations per family: E-DDQN 2,985,984, DQN 497,664, E-CB 248,832, CS-MLP 4,608, HGB 1,728, random forest 192, logistic regression 24. Each family received 16 random configurations (12 for logistic regression), so the budget is matched in number, not in coverage.

S2.2 Selected configuration per outer fold

Selection criterion: pooled inner out-of-fold 5:1 expected cost at the cost-minimising threshold (ties: PR-AUC).

Model	Outer	OOF cost	Selected configuration
E-DDQN	0	0.424	feature_set=exposure+profile, hidden=[128, 128], gamma=0.99, full_info=True, rho=0.4, tau=0.05, norm=layer, dropout=0.2, lr=0.003, weight_decay=0.0, batch_size=256, eps_decay=0.97, warmup=0, updates_per_learn=4
E-DDQN	1	0.452	feature_set=exposure+profile, hidden=[128, 128], gamma=0.99, full_info=True, rho=0.4, tau=0.05, norm=layer, dropout=0.2, lr=0.003, weight_decay=0.0, batch_size=256, eps_decay=0.97, warmup=0, updates_per_learn=4
E-DDQN	2	0.484	feature_set=exposure+profile, hidden=[128, 128], gamma=0.99, full_info=True, rho=0.4, tau=0.05, norm=layer, dropout=0.2, lr=0.003, weight_decay=0.0, batch_size=256, eps_decay=0.97, warmup=0, updates_per_learn=4

Model	Outer	OOF cost	Selected configuration
E-DDQN	3	0.490	feature_set=exposure+profile, hidden=[128, 128], gamma=0.99, full_info=False, rho=0.4, tau=0.005, norm=layer, dropout=0.3, lr=0.0003, weight_decay=0.0001, batch_size=512, eps_decay=0.9, warmup=0, updates_per_learn=1
E-DDQN	4	0.483	feature_set=exposure+profile, hidden=[256, 512, 256], gamma=0.9, full_info=False, rho=0.5, tau=0.05, norm=layer, dropout=0.3, lr=0.0003, weight_decay=0.001, batch_size=512, eps_decay=0.95, warmup=4000, updates_per_learn=2
E-CB	0	0.472	feature_set=exposure, hidden=[256, 512, 256], gamma=0.0, full_info=True, rho=0.5, norm=layer, dropout=0.3, lr=0.003, weight_decay=0.001, batch_size=512, eps_decay=0.97, warmup=4000, updates_per_learn=4
E-CB	1	0.459	feature_set=exposure, hidden=[256, 512, 256], gamma=0.0, full_info=True, rho=0.5, norm=layer, dropout=0.3, lr=0.003, weight_decay=0.001, batch_size=512, eps_decay=0.97, warmup=4000, updates_per_learn=4
E-CB	2	0.518	feature_set=exposure, hidden=[256, 512, 256], gamma=0.0, full_info=True, rho=0.5, norm=layer, dropout=0.3, lr=0.003, weight_decay=0.001, batch_size=512, eps_decay=0.97, warmup=4000, updates_per_learn=4
E-CB	3	0.523	feature_set=exposure, hidden=[128, 128], gamma=0.0, full_info=False, rho=0.6, norm=layer, dropout=0.2, lr=0.001, weight_decay=0.001, batch_size=256, eps_decay=0.97, warmup=0, updates_per_learn=2
E-CB	4	0.466	feature_set=exposure+profile, hidden=[64, 64], gamma=0.0, full_info=True, rho=0.5, norm=layer, dropout=0.2, lr=0.003, weight_decay=0.001, batch_size=256, eps_decay=0.95, warmup=0, updates_per_learn=1
DQN	0	0.432	feature_set=exposure, hidden=[64, 64], gamma=0.0, full_info=False, hard_every=10, norm=layer, dropout=0.0, lr=0.001, weight_decay=0.0001, batch_size=256, eps_decay=0.95, warmup=0, updates_per_learn=2
DQN	1	0.452	feature_set=exposure+profile, hidden=[256, 512, 256], gamma=0.9, full_info=False, hard_every=500, norm=layer, dropout=0.0, lr=0.001, weight_decay=0.001, batch_size=512, eps_decay=0.9, warmup=0, updates_per_learn=4
DQN	2	0.538	feature_set=exposure, hidden=[128, 128], gamma=0.9, full_info=False, hard_every=500, norm=batch, dropout=0.3, lr=0.0003, weight_decay=0.0, batch_size=256, eps_decay=0.95, warmup=0, updates_per_learn=2
DQN	3	0.501	feature_set=exposure, hidden=[128, 128], gamma=0.9, full_info=False, hard_every=100, norm=layer, dropout=0.2, lr=0.001, weight_decay=0.0, batch_size=512, eps_decay=0.97, warmup=0, updates_per_learn=2
DQN	4	0.469	feature_set=exposure+profile, hidden=[256, 512, 256], gamma=0.0, full_info=False, hard_every=100, norm=batch, dropout=0.1, lr=0.003, weight_decay=0.0, batch_size=512, eps_decay=0.97, warmup=4000, updates_per_learn=1
CS-MLP	0	0.485	feature_set=exposure+profile, hidden=[128, 128], norm=layer, dropout=0.3, lr=0.001, weight_decay=0.0, batch_size=256, sampling=balanced
CS-MLP	1	0.490	feature_set=exposure+profile, hidden=[128, 128], norm=layer, dropout=0.3, lr=0.001, weight_decay=0.0, batch_size=256, sampling=balanced
CS-MLP	2	0.503	feature_set=exposure, hidden=[64, 64], norm=layer, dropout=0.3, lr=0.0003, weight_decay=0.0, batch_size=256, sampling=uniform
CS-MLP	3	0.527	feature_set=telemetry, hidden=[64, 64], norm=layer, dropout=0.0, lr=0.003, weight_decay=0.0001, batch_size=256, sampling=balanced
CS-MLP	4	0.500	feature_set=telemetry, hidden=[64, 64], norm=layer, dropout=0.3, lr=0.0003, weight_decay=0.001, batch_size=512, sampling=uniform
LogReg	0	0.486	feature_set=exposure, C=0.0001
LogReg	1	0.522	feature_set=telemetry+profile, C=1.0
LogReg	2	0.540	feature_set=exposure, C=10.0
LogReg	3	0.511	feature_set=exposure, C=0.0001
LogReg	4	0.531	feature_set=exposure+profile, C=0.01
HGB	0	0.486	feature_set=exposure, learning_rate=0.02, max_iter=50, max_leaf_nodes=4, min_samples_leaf=20, l2_regularization=1.0
HGB	1	0.501	feature_set=exposure, learning_rate=0.02, max_iter=50, max_leaf_nodes=4, min_samples_leaf=20, l2_regularization=1.0
HGB	2	0.542	feature_set=exposure, learning_rate=0.02, max_iter=400, max_leaf_nodes=4, min_samples_leaf=20, l2_regularization=0.0
HGB	3	0.532	feature_set=telemetry, learning_rate=0.05, max_iter=200, max_leaf_nodes=31, min_samples_leaf=100, l2_regularization=0.0
HGB	4	0.527	feature_set=exposure, learning_rate=0.02, max_iter=400, max_leaf_nodes=4, min_samples_leaf=20, l2_regularization=0.0

Model	Outer	OOF cost	Selected configuration
RF	0	0.465	feature_set=telemetry, min_samples_leaf=100, max_depth=3, max_features=0.5
RF	1	0.478	feature_set=exposure, min_samples_leaf=25, max_depth=8, max_features=sqrt
RF	2	0.509	feature_set=exposure, min_samples_leaf=5, max_depth=5, max_features=sqrt
RF	3	0.514	feature_set=exposure, min_samples_leaf=5, max_depth=5, max_features=sqrt
RF	4	0.510	feature_set=telemetry+profile, min_samples_leaf=25, max_depth=3, max_features=sqrt

S3 Additional results

S3.1 Full metric panel (primary rule D1)

Mean ± SD over five seeds (%, except MCC and cost); TP/TN/FP/FN are the exact confusion-matrix counts of the seed-ensemble predictions on all 9,390 outer-test samples. FPR = 1−specificity and FNR = 1−recall.

Model	Acc.	Prec.	Rec.	Spec.	F1	Bal.acc.	MCC	G-mean	FPR	FNR	$\text{Cost}_{5:1}$	TP	TN	FP	FN
E-DDQN	45.4±1.3	44.4±0.7	93.4±2.5	7.4±2.7	60.2±0.9	50.4±1.2	0.016±0.054	26.0±4.5	92.6±2.7	6.6±2.5	0.663±0.048	3,747	423	4,820	400
E-CB	47.6±1.2	45.6±0.6	97.1±2.9	8.4±2.8	62.1±1.0	52.8±1.2	0.122±0.052	28.2±5.0	91.6±2.8	2.9±2.9	0.575±0.055	4,007	763	4,480	140
DQN	50.1±1.9	46.8±1.0	95.2±2.7	14.5±4.9	62.8±0.8	54.8±1.6	0.158±0.038	36.6±5.8	85.5±4.9	4.8±2.7	0.584±0.040	4,018	852	4,391	129
CS-MLP	45.4±0.9	44.7±0.4	99.4±0.7	2.8±1.9	61.7±0.4	51.1±0.8	0.072±0.049	14.7±8.8	97.2±1.9	0.6±0.7	0.557±0.012	4,124	318	4,925	23
LogReg	48.3±0.0	46.0±0.0	99.1±0.0	8.1±0.0	62.9±0.0	53.6±0.0	0.167±0.000	28.4±0.0	91.9±0.0	0.9±0.0	0.532±0.000	4,111	426	4,817	36
HGB	51.0±0.0	47.3±0.0	97.0±0.0	14.6±0.0	63.6±0.0	55.8±0.0	0.197±0.000	37.6±0.0	85.4±0.0	3.0±0.0	0.543±0.000	4,023	764	4,479	124
RF	53.6±0.3	48.8±0.2	98.5±0.3	18.2±0.7	65.2±0.1	58.3±0.3	0.266±0.004	42.3±0.8	81.8±0.7	1.5±0.3	0.491±0.003	4,093	962	4,281	54

S3.2 Secondary operating rules

Mean ± SD over five seeds; FN and FP are the exact counts of the seed ensemble, as in the main text. In the two tables below, the name of the proposed model and the lowest cost among the primary models are set in bold. Rule D0 (untuned cut-off $\hat{p} \geq 1/6$, the nominal 5:1 Bayes cut-off; for the asymmetric agents the greedy policy; its effective position on $P(y = 1 \mid s)$ for each model is given in Section S5.12):

Model	Recall	Spec.	Prec.	F1	Bal. acc.	MCC	$\text{Cost}_{5:1}$	ROC-AUC	PR-AUC	FN	FP
E-DDQN	77.2±3.8	28.0±2.0	45.9±1.7	57.6±2.4	52.6±2.7	0.060±0.061	0.905±0.092	0.538±0.016	0.483±0.019	909	3,862
E-CB	85.9±2.2	37.2±3.0	52.0±1.1	64.7±1.1	61.5±1.5	0.258±0.031	0.663±0.044	0.653±0.011	0.562±0.006	627	3,262
DQN	85.1±1.7	37.1±1.5	51.7±0.7	64.3±0.9	61.1±1.0	0.248±0.022	0.680±0.035	0.661±0.013	0.595±0.022	561	3,330
CS-MLP	83.0±3.1	36.4±2.3	50.8±1.0	63.0±1.5	59.7±1.6	0.215±0.037	0.731±0.066	0.644±0.018	0.521±0.022	622	3,381
LogReg	85.3±0.0	13.6±0.0	43.8±0.0	57.9±0.0	49.4±0.0	−0.016±0.000	0.808±0.000	0.626±0.000	0.576±0.000	611	4,531
HGB	92.0±0.0	30.0±0.0	51.0±0.0	65.6±0.0	61.0±0.0	0.272±0.000	0.567±0.000	0.692±0.000	0.593±0.000	331	3,670
RF	94.5±0.1	34.9±0.1	53.4±0.0	68.3±0.0	64.7±0.0	0.353±0.001	**0.486±0.001**	0.688±0.002	0.548±0.001	228	3,410
Always alarm	100.0	0.0	44.2	61.3	50.0	0.000	0.558	–	–	0	5,243
Never alarm	0.0	100.0	–	–	50.0	0.000	2.208	–	–	4,147	0

Validation recall constraint (≥0.95 on pooled inner-OOF predictions):

Model	Recall	Spec.	Prec.	F1	Bal. acc.	MCC	$\text{Cost}_{5:1}$	ROC-AUC	PR-AUC	FN	FP
E-DDQN	86.8±3.1	11.5±3.4	43.7±0.9	58.1±1.3	49.1±1.7	−0.027±0.053	0.785±0.062	0.538±0.016	0.483±0.019	641	4,539
E-CB	95.2±2.7	15.0±3.3	47.0±0.8	62.9±1.0	55.1±1.4	0.167±0.042	0.581±0.051	0.653±0.011	0.562±0.006	317	4,093
DQN	93.5±2.9	23.0±3.4	49.0±0.8	64.3±1.0	58.3±1.2	0.227±0.032	0.573±0.052	0.661±0.013	0.595±0.022	343	3,810
CS-MLP	98.3±1.2	9.3±4.2	46.2±1.0	62.8±0.9	53.8±1.9	0.156±0.052	0.544±0.025	0.644±0.018	0.521±0.022	105	4,463
LogReg	97.2±0.0	18.3±0.0	48.5±0.0	64.7±0.0	57.7±0.0	0.241±0.000	0.518±0.000	0.626±0.000	0.576±0.000	116	4,285
HGB	97.5±0.0	18.3±0.0	48.6±0.0	64.8±0.0	57.9±0.0	0.248±0.000	0.511±0.000	0.692±0.000	0.593±0.000	103	4,282
RF	97.0±0.2	25.4±0.3	50.7±0.0	66.6±0.0	61.2±0.0	0.307±0.002	**0.483±0.004**	0.688±0.002	0.548±0.001	128	3,896
Always alarm	100.0	0.0	44.2	61.3	50.0	0.000	0.558	–	–	0	5,243
Never alarm	0.0	100.0	–	–	50.0	0.000	2.208	–	–	4,147	0

S3.3 Group-bootstrap confidence intervals

Seed-ensemble predictions, rule D1; 95 % percentile intervals from 2,000 bootstrap replicates of the 21 linkage groups.

Model (seed ensemble, D1)	Recall	Spec.	$\text{Cost}_{5:1}$	ROC-AUC	PR-AUC
E-DDQN	90.4 [77.7, 98.9]	8.1 [1.5, 17.6]	0.726 [0.533, 0.996]	0.541 [0.382, 0.690]	0.496 [0.329, 0.683]
E-CB	96.6 [92.2, 99.6]	14.6 [8.3, 21.4]	0.552 [0.431, 0.673]	0.659 [0.550, 0.767]	0.565 [0.429, 0.740]
DQN	96.9 [92.6, 99.7]	16.3 [8.7, 24.8]	0.536 [0.417, 0.654]	0.665 [0.558, 0.785]	0.606 [0.456, 0.778]
CS-MLP	99.4 [98.5, 100.0]	6.1 [2.6, 10.8]	0.537 [0.421, 0.634]	0.654 [0.527, 0.781]	0.529 [0.393, 0.712]
LogReg	99.1 [98.0, 100.0]	8.1 [4.0, 13.0]	0.532 [0.429, 0.627]	0.626 [0.498, 0.750]	0.576 [0.408, 0.754]
HGB	97.0 [92.4, 99.9]	14.6 [7.0, 24.9]	0.543 [0.439, 0.654]	0.692 [0.574, 0.809]	0.593 [0.437, 0.779]
RF	98.7 [97.5, 99.6]	18.3 [11.7, 28.0]	0.485 [0.381, 0.577]	0.688 [0.582, 0.808]	0.548 [0.443, 0.735]

S3.4 Per-fold results

Expected 5:1 cost / ROC-AUC of the seed ensemble in each outer fold (rule D1). Fold-to-fold variation is large because each fold contains only 3–5 linkage groups.

Model	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4
E-DDQN	0.604 / 0.536	0.762 / 0.522	0.529 / 0.734	1.184 / 0.501	0.693 / 0.466
E-CB	0.569 / 0.655	0.537 / 0.718	0.462 / 0.756	0.460 / 0.747	0.758 / 0.447
DQN	0.564 / 0.715	0.757 / 0.521	0.451 / 0.852	0.426 / 0.740	0.520 / 0.499
CS-MLP	0.604 / 0.575	0.542 / 0.596	0.495 / 0.820	0.528 / 0.708	0.501 / 0.659
LogReg	0.582 / 0.623	0.542 / 0.428	0.486 / 0.810	0.531 / 0.712	0.515 / 0.610
HGB	0.619 / 0.631	0.547 / 0.740	0.510 / 0.791	0.522 / 0.761	0.495 / 0.585
RF	0.600 / 0.630	0.463 / 0.728	0.397 / 0.771	0.457 / 0.761	0.486 / 0.685

S3.5 Event-level behavior

Seed ensembles at rule D1. Alarm percentages are the fraction of seconds with an alarm in each session, averaged over sessions (before onset: the 26 sessions with an event; event-free: the 9 sessions without one).

Model	Events detected	Median delay (s)	Alarm % before onset	Alarm % in event-free sessions
E-DDQN	26/26	0	82.3	96.6
E-CB	25/26	0	73.1	76.9
DQN	25/26	0	77.4	70.4
CS-MLP	26/26	0	92.9	88.1
LogReg	25/26	0	76.0	91.7
HGB	25/26	0	69.3	90.8
RF	25/26	0	61.3	75.3

S3.6 Paired tests under the secondary rules

Rule D0:

B (vs. A = E-DDQN)	b	c	err_A	err_B	χ^2_{cc}	p_{exact}	$\Delta\mathrm{Cost}_{5:1}$ (A−B) [95% CI]	p_{group}
E-CB	635	1,517	4,771	3,889	360.7	<0.001	+0.214 [+0.025, +0.487]	0.018
DQN	467	1,347	4,771	3,891	425.9	<0.001	+0.242 [+0.039, +0.542]	0.012
CS-MLP	431	1,199	4,771	4,003	360.9	<0.001	+0.204 [−0.030, +0.507]	0.11
LogReg	1,297	926	4,771	5,142	61.6	<0.001	+0.087 [−0.293, +0.541]	0.69
HGB	833	1,603	4,771	4,001	242.8	<0.001	+0.328 [+0.078, +0.639]	0.003
RF	526	1,659	4,771	3,638	586.5	<0.001	+0.411 [+0.147, +0.748]	0.001

Validation recall ≥ 0.95:

B (vs. A = E-DDQN)	b	c	err_A	err_B	χ^2_{cc}	p_{exact}	$\Delta\mathrm{Cost}_{5:1}$ (A−B) [95% CI]	p_{group}
E-CB	378	1,148	5,180	4,410	387.5	<0.001	+0.220 [+0.017, +0.566]	0.021
DQN	294	1,321	5,180	4,153	651.8	<0.001	+0.236 [+0.013, +0.580]	0.028
CS-MLP	509	1,121	5,180	4,568	229.0	<0.001	+0.294 [+0.033, +0.652]	0.013
LogReg	538	1,317	5,180	4,401	326.3	<0.001	+0.307 [+0.027, +0.687]	0.02
HGB	434	1,229	5,180	4,385	379.1	<0.001	+0.314 [+0.030, +0.711]	0.023
RF	393	1,549	5,180	4,024	686.9	<0.001	+0.342 [+0.067, +0.719]	0.003

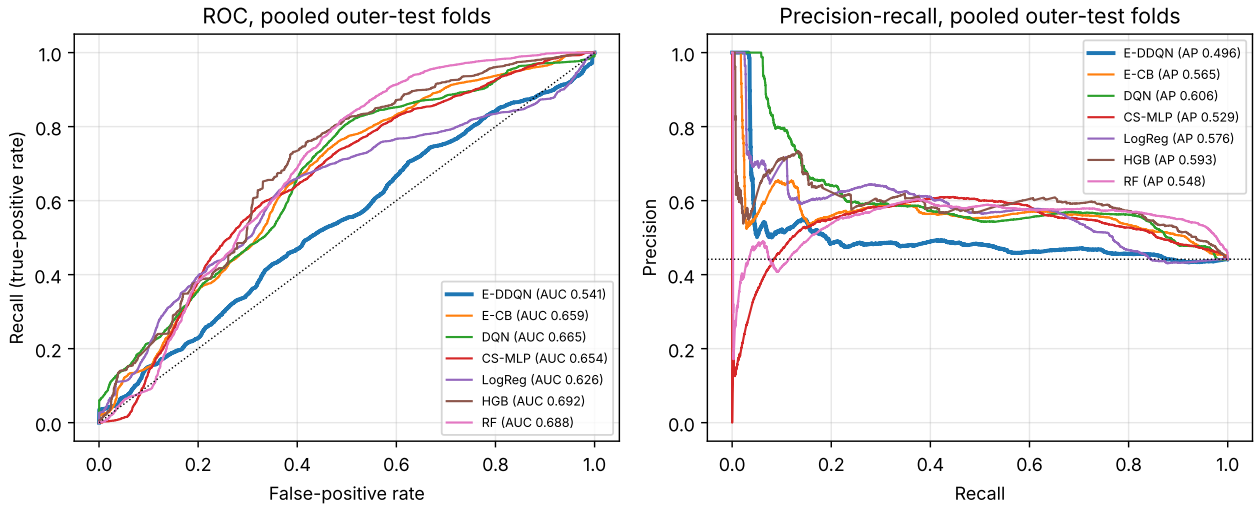
S3.7 Value-estimation diagnostics

Mean Q-value of the greedy action on the inner-validation groups compared with the discounted return actually obtained by the greedy policy on those sessions (rewards of every action are known offline). For E-DDQN and E-CB, the comparison is also affected by the class mix of the class-balanced replay, which differs from that of the validation data (Section S5.12).

Model	γ (selected)	mean $Q(s, \pi(s))$	mean MC return	bias (mean ± SD over runs)
DQN	0–0.9	1.467	1.074	+0.393 ± 1.493
E-CB	0–0	0.224	0.136	+0.088 ± 0.078
E-DDQN	0.9–0.99	5.998	8.898	−2.900 ± 8.250

S3.8 ROC and precision–recall curves (primary models)

Seed-ensemble scores pooled over the five outer-test folds; AP in the legend is the average precision, i.e. the PR-AUC of the tables.



S4 Telemetry-perturbation stress tests

S4.1 Design (frozen before evaluation; `configs/attacks.json`)

Perturbations act on the raw per-second channels of the outer-test sessions; causal features are then recomputed with the training-fitted featurizer and scaler of each inner model. Cybersickness labels are never modified; the indicator z marks perturbed seconds. Stress scenarios (each over the whole session or one random 60-s window): frame-rate cap at 72, 60 or 45 fps (Type I); uniform orientation jitter of up to ± 0.02, 0.05 or 0.10 per rotation component (Type II); speed up-scaling by 10, 25 or 50 % with acceleration recomputed as the per-second speed difference (the logged acceleration equals this difference in every session; Type II); speed raised by 10, 20 or 40 units for one second every 20 s (Type II). Together: 4 kinds $\times$ 3 strengths $\times$ 2 windows $=$ 24 stress variants. Evasion-only candidates (whole session): speed down-scaling by 25 or 50 %, orientation smoothing with $\lambda = 0.5$ or 0.25, frame-rate restoration to 90 fps (5 variants; 29 variants in total, 17 of them whole-session). Speed is clipped to each game's observed range and orientation components to their valid range $[-1, 1]$. The frame-rate caps are applied as caps: the 60- and 45-fps caps lie below every frame rate observed in Flight sessions after the first second (70–94 fps), whereas the Race sessions range from 36 to 92 fps, so these scenarios are outside the data range for Flight. Field-of-view manipulation is not evaluated because the field of view never varies within a game in the data.

S4.2 Interpretation limits

The tests measure how bounded manipulation of the logged channels moves the detectors' scores and decisions. They do not show that a manipulation would cause or hide cybersickness in a person, because no physiological effect was measured. The adaptive worst cases give the adversary the labels and choose the most damaging candidate per session and model.

S4.3 Operating points and adaptive worst cases

Recall/false-positive rate (%) of the seed ensembles on perturbed seconds at the D1 operating point for representative whole-session variants, and per-session adaptive worst cases (evasion: minimum recall; false-alarm induction: maximum FPR) over all 17 whole-session candidates.

Model	Clean Rec./FPR	fps≤45 Rec./FPR	jitter 0.10 Rec./FPR	speed ×1.5 Rec./FPR	spikes 40 Rec./FPR	Adaptive worst case Rec. (evasion)	Adaptive worst case FPR (false alarm)
E-DDQN	90.4/91.9	90.4/91.9	94.0/92.2	93.5/90.7	95.4/94.3	82.9	94.5
E-CB	96.6/85.4	96.6/85.4	96.1/88.2	98.7/95.9	99.3/97.7	78.0	98.9
DQN	96.9/83.7	96.9/83.7	94.8/83.5	97.8/91.8	97.9/95.4	87.3	97.5
CS-MLP	99.4/93.9	100.0/98.3	99.7/96.5	100.0/98.2	99.9/99.1	95.6	100.0
LogReg	99.1/91.9	97.0/86.3	99.2/92.7	99.7/94.9	99.9/96.7	95.4	96.7
HGB	97.0/85.4	97.0/85.9	96.3/82.7	98.2/91.7	98.1/95.7	81.0	96.1
RF	98.7/81.7	98.7/81.3	98.0/81.1	99.0/92.1	99.8/97.4	90.5	98.4

S4.4 All variants

Recall and false-positive rate on perturbed seconds (%) and 5:1 cost on those seconds, clean vs. perturbed, seed ensembles at the D1 operating point.

Model	Scenario	Value	Window	Rec. clean	Rec. att.	FPR clean	FPR att.	Cost clean	Cost att.
CS-MLP	accel spikes	10	whole	99.4	99.6	93.9	95.5	0.537	0.542
CS-MLP	accel spikes	10	window60	99.3	99.7	90.5	92.4	0.532	0.536
CS-MLP	accel spikes	20	whole	99.4	99.8	93.9	98.3	0.537	0.553
CS-MLP	accel spikes	20	window60	100.0	100.0	89.2	97.9	0.545	0.599
CS-MLP	accel spikes	40	whole	99.4	99.9	93.9	99.1	0.537	0.556
CS-MLP	accel spikes	40	window60	100.0	100.0	97.0	99.6	0.496	0.510
CS-MLP	fps cap	45	whole	99.4	100.0	93.9	98.3	0.537	0.549
CS-MLP	fps cap	45	window60	99.5	100.0	95.4	98.8	0.509	0.514
CS-MLP	fps cap	60	whole	99.4	100.0	93.9	98.3	0.537	0.549
CS-MLP	fps cap	60	window60	98.6	100.0	95.4	99.0	0.649	0.649
CS-MLP	fps cap	72	whole	99.4	100.0	93.9	98.3	0.537	0.549
CS-MLP	fps cap	72	window60	99.5	100.0	96.6	98.8	0.511	0.510
CS-MLP	fps restore	90	whole	99.4	99.7	93.9	93.7	0.537	0.530
CS-MLP	rot jitter	0.02	whole	99.4	99.1	93.9	92.0	0.537	0.534
CS-MLP	rot jitter	0.02	window60	99.9	99.7	96.7	94.6	0.517	0.511
CS-MLP	rot jitter	0.05	whole	99.4	98.8	93.9	91.4	0.537	0.536
CS-MLP	rot jitter	0.05	window60	98.3	96.4	94.4	91.2	0.656	0.667
CS-MLP	rot jitter	0.1	whole	99.4	99.7	93.9	96.5	0.537	0.545
CS-MLP	rot jitter	0.1	window60	100.0	99.8	94.3	96.2	0.560	0.576
CS-MLP	rot smooth	0.25	whole	99.4	99.5	93.9	97.1	0.537	0.553
CS-MLP	rot smooth	0.5	whole	99.4	99.3	93.9	96.8	0.537	0.555
CS-MLP	speed scale down	−0.5	whole	99.4	95.6	93.9	85.3	0.537	0.573
CS-MLP	speed scale down	−0.25	whole	99.4	98.5	93.9	90.2	0.537	0.538
CS-MLP	speed scale up	0.1	whole	99.4	99.6	93.9	95.1	0.537	0.540
CS-MLP	speed scale up	0.1	window60	99.3	99.7	92.2	94.1	0.541	0.544
CS-MLP	speed scale up	0.25	whole	99.4	100.0	93.9	96.7	0.537	0.541
CS-MLP	speed scale up	0.25	window60	100.0	100.0	95.3	97.2	0.534	0.544
CS-MLP	speed scale up	0.5	whole	99.4	100.0	93.9	98.2	0.537	0.548
CS-MLP	speed scale up	0.5	window60	100.0	100.0	87.4	97.1	0.448	0.497
DQN	accel spikes	10	whole	96.9	96.8	83.7	86.1	0.536	0.552
DQN	accel spikes	10	window60	92.6	92.6	80.3	82.7	0.617	0.631
DQN	accel spikes	20	whole	96.9	98.3	83.7	95.5	0.536	0.570
DQN	accel spikes	20	window60	96.9	97.4	78.5	98.0	0.540	0.650
DQN	accel spikes	40	whole	96.9	97.9	83.7	95.4	0.536	0.578
DQN	accel spikes	40	window60	99.5	100.0	87.5	99.8	0.460	0.511
DQN	fps cap	45	whole	96.9	96.9	83.7	83.7	0.536	0.536
DQN	fps cap	45	window60	94.1	94.1	84.5	84.5	0.581	0.581
DQN	fps cap	60	whole	96.9	96.9	83.7	83.7	0.536	0.536
DQN	fps cap	60	window60	94.5	94.5	83.7	83.7	0.643	0.643
DQN	fps cap	72	whole	96.9	96.9	83.7	83.7	0.536	0.536
DQN	fps cap	72	window60	93.5	93.5	83.9	83.9	0.591	0.591
DQN	fps restore	90	whole	96.9	96.9	83.7	83.7	0.536	0.536
DQN	rot jitter	0.02	whole	96.9	94.2	83.7	78.0	0.536	0.563
DQN	rot jitter	0.02	window60	95.6	95.5	85.1	83.4	0.555	0.548
DQN	rot jitter	0.05	whole	96.9	93.2	83.7	74.7	0.536	0.568
DQN	rot jitter	0.05	window60	92.5	78.9	82.7	74.8	0.676	0.852
DQN	rot jitter	0.1	whole	96.9	94.8	83.7	83.5	0.536	0.581
DQN	rot jitter	0.1	window60	96.3	95.7	86.6	84.7	0.590	0.591
DQN	rot smooth	0.25	whole	96.9	99.6	83.7	92.4	0.536	0.525
DQN	rot smooth	0.5	whole	96.9	98.9	83.7	91.0	0.536	0.533
DQN	speed scale down	−0.5	whole	96.9	92.2	83.7	70.6	0.536	0.567
DQN	speed scale down	−0.25	whole	96.9	94.7	83.7	76.9	0.536	0.547
DQN	speed scale up	0.1	whole	96.9	97.0	83.7	88.1	0.536	0.557
DQN	speed scale up	0.1	window60	92.6	92.6	84.6	86.5	0.640	0.651
DQN	speed scale up	0.25	whole	96.9	97.4	83.7	90.8	0.536	0.564
DQN	speed scale up	0.25	window60	97.2	97.0	82.6	87.9	0.523	0.558
DQN	speed scale up	0.5	whole	96.9	97.8	83.7	91.8	0.536	0.562
DQN	speed scale up	0.5	window60	99.5	99.5	76.4	86.4	0.403	0.455
E-CB	accel spikes	10	whole	96.6	97.9	85.4	88.3	0.552	0.540
E-CB	accel spikes	10	window60	97.0	98.2	77.4	79.1	0.506	0.491
E-CB	accel spikes	20	whole	96.6	98.9	85.4	97.0	0.552	0.566
E-CB	accel spikes	20	window60	95.0	96.0	76.4	94.8	0.564	0.657
E-CB	accel spikes	40	whole	96.6	99.3	85.4	97.7	0.552	0.562
E-CB	accel spikes	40	window60	99.5	100.0	89.4	99.3	0.469	0.508
E-CB	fps cap	45	whole	96.6	96.6	85.4	85.4	0.552	0.552
E-CB	fps cap	45	window60	96.0	96.0	84.2	84.2	0.533	0.533
E-CB	fps cap	60	whole	96.6	96.6	85.4	85.4	0.552	0.552
E-CB	fps cap	60	window60	94.6	94.6	87.2	87.2	0.663	0.663
E-CB	fps cap	72	whole	96.6	96.6	85.4	85.4	0.552	0.552
E-CB	fps cap	72	window60	97.0	97.0	86.2	86.2	0.518	0.518
E-CB	fps restore	90	whole	96.6	96.6	85.4	85.4	0.552	0.552
E-CB	rot jitter	0.02	whole	96.6	96.5	85.4	76.3	0.552	0.504
E-CB	rot jitter	0.02	window60	97.9	95.5	85.3	81.9	0.502	0.540
E-CB	rot jitter	0.05	whole	96.6	94.4	85.4	79.1	0.552	0.565
E-CB	rot jitter	0.05	window60	93.9	86.8	84.6	71.4	0.664	0.695
E-CB	rot jitter	0.1	whole	96.6	96.1	85.4	88.2	0.552	0.578
E-CB	rot jitter	0.1	window60	97.7	95.0	88.5	82.7	0.572	0.593
E-CB	rot smooth	0.25	whole	96.6	95.9	85.4	94.4	0.552	0.618
E-CB	rot smooth	0.5	whole	96.6	95.3	85.4	90.9	0.552	0.612
E-CB	speed scale down	−0.5	whole	96.6	81.0	85.4	59.5	0.552	0.752
E-CB	speed scale down	−0.25	whole	96.6	88.9	85.4	71.2	0.552	0.643
E-CB	speed scale up	0.1	whole	96.6	98.0	85.4	91.0	0.552	0.552

Model	Scenario	Value	Window	Rec. clean	Rec. att.	FPR clean	FPR att.	Cost clean	Cost att.
E-CB	speed scale up	0.1	window60	97.4	98.2	79.4	82.2	0.509	0.508
E-CB	speed scale up	0.25	whole	96.6	98.2	85.4	94.1	0.552	0.565
E-CB	speed scale up	0.25	window60	95.6	95.4	84.8	91.4	0.572	0.614
E-CB	speed scale up	0.5	whole	96.6	98.7	85.4	95.9	0.552	0.564
E-CB	speed scale up	0.5	window60	99.1	99.5	73.3	89.3	0.397	0.469
E-DDQN	accel spikes	10	whole	90.4	90.5	91.9	92.1	0.726	0.723
E-DDQN	accel spikes	10	window60	91.7	91.4	97.9	98.0	0.737	0.745
E-DDQN	accel spikes	20	whole	90.4	94.9	91.9	93.8	0.726	0.637
E-DDQN	accel spikes	20	window60	88.8	90.4	94.5	96.0	0.796	0.774
E-DDQN	accel spikes	40	whole	90.4	95.4	91.9	94.3	0.726	0.627
E-DDQN	accel spikes	40	window60	94.0	94.0	98.7	99.9	0.651	0.657
E-DDQN	fps cap	45	whole	90.4	90.4	91.9	91.9	0.726	0.726
E-DDQN	fps cap	45	window60	90.2	90.2	99.3	99.3	0.752	0.752
E-DDQN	fps cap	60	whole	90.4	90.4	91.9	91.9	0.726	0.726
E-DDQN	fps cap	60	window60	83.7	83.7	95.5	95.5	0.907	0.907
E-DDQN	fps cap	72	whole	90.4	90.4	91.9	91.9	0.726	0.726
E-DDQN	fps cap	72	window60	89.1	89.1	99.2	99.2	0.775	0.775
E-DDQN	fps restore	90	whole	90.4	90.4	91.9	91.9	0.726	0.726
E-DDQN	rot jitter	0.02	whole	90.4	92.7	91.9	91.4	0.726	0.671
E-DDQN	rot jitter	0.02	window60	91.4	92.7	100.0	100.0	0.734	0.702
E-DDQN	rot jitter	0.05	whole	90.4	93.9	91.9	91.2	0.726	0.644
E-DDQN	rot jitter	0.05	window60	83.2	82.1	88.5	88.5	0.870	0.889
E-DDQN	rot jitter	0.1	whole	90.4	94.0	91.9	92.2	0.726	0.648
E-DDQN	rot jitter	0.1	window60	91.0	84.7	88.5	91.4	0.708	0.854
E-DDQN	rot smooth	0.25	whole	90.4	88.0	91.9	91.8	0.726	0.777
E-DDQN	rot smooth	0.5	whole	90.4	88.9	91.9	92.3	0.726	0.760
E-DDQN	speed scale down	−0.5	whole	90.4	86.7	91.9	90.9	0.726	0.802
E-DDQN	speed scale down	−0.25	whole	90.4	88.2	91.9	91.7	0.726	0.772
E-DDQN	speed scale up	0.1	whole	90.4	91.4	91.9	92.0	0.726	0.704
E-DDQN	speed scale up	0.1	window60	91.3	92.0	98.4	98.4	0.748	0.734
E-DDQN	speed scale up	0.25	whole	90.4	92.8	91.9	91.8	0.726	0.672
E-DDQN	speed scale up	0.25	window60	87.0	88.3	99.8	99.9	0.845	0.817
E-DDQN	speed scale up	0.5	whole	90.4	93.5	91.9	90.7	0.726	0.650
E-DDQN	speed scale up	0.5	window60	94.0	94.0	91.0	90.9	0.612	0.611
HGB	accel spikes	10	whole	97.0	97.1	85.4	87.8	0.543	0.555
HGB	accel spikes	10	window60	99.3	99.3	86.1	88.5	0.507	0.521
HGB	accel spikes	20	whole	97.0	97.9	85.4	94.5	0.543	0.575
HGB	accel spikes	20	window60	91.2	92.8	81.8	95.3	0.670	0.724
HGB	accel spikes	40	whole	97.0	98.1	85.4	95.7	0.543	0.577
HGB	accel spikes	40	window60	99.5	100.0	90.1	97.6	0.473	0.500
HGB	fps cap	45	whole	97.0	97.0	85.4	85.9	0.543	0.545
HGB	fps cap	45	window60	93.9	93.9	89.0	89.2	0.609	0.610
HGB	fps cap	60	whole	97.0	97.0	85.4	85.9	0.543	0.545
HGB	fps cap	60	window60	97.5	97.5	92.3	92.4	0.648	0.649
HGB	fps cap	72	whole	97.0	97.0	85.4	85.9	0.543	0.545
HGB	fps cap	72	window60	99.4	99.4	88.4	88.6	0.471	0.472
HGB	fps restore	90	whole	97.0	97.0	85.4	86.9	0.543	0.551
HGB	rot jitter	0.02	whole	97.0	97.0	85.4	85.0	0.543	0.541
HGB	rot jitter	0.02	window60	99.4	99.4	87.9	87.7	0.482	0.481
HGB	rot jitter	0.05	whole	97.0	96.6	85.4	82.9	0.543	0.538
HGB	rot jitter	0.05	window60	91.6	90.2	84.6	82.0	0.703	0.710
HGB	rot jitter	0.1	whole	97.0	96.3	85.4	82.7	0.543	0.543
HGB	rot jitter	0.1	window60	91.3	89.9	84.0	77.4	0.676	0.664
HGB	rot smooth	0.25	whole	97.0	97.0	85.4	88.3	0.543	0.558
HGB	rot smooth	0.5	whole	97.0	96.9	85.4	88.0	0.543	0.559
HGB	speed scale down	−0.5	whole	97.0	81.0	85.4	64.4	0.543	0.780
HGB	speed scale down	−0.25	whole	97.0	91.0	85.4	78.3	0.543	0.635
HGB	speed scale up	0.1	whole	97.0	97.3	85.4	88.9	0.543	0.556
HGB	speed scale up	0.1	window60	99.4	99.4	81.9	83.3	0.479	0.487
HGB	speed scale up	0.25	whole	97.0	98.0	85.4	90.7	0.543	0.550
HGB	speed scale up	0.25	window60	92.9	94.3	88.1	92.3	0.649	0.643
HGB	speed scale up	0.5	whole	97.0	98.2	85.4	91.7	0.543	0.551
HGB	speed scale up	0.5	window60	99.4	99.4	77.3	87.1	0.411	0.461
LogReg	accel spikes	10	whole	99.1	99.2	91.9	92.6	0.532	0.536
LogReg	accel spikes	10	window60	99.4	99.4	88.9	89.5	0.520	0.524
LogReg	accel spikes	20	whole	99.1	99.3	91.9	94.8	0.532	0.545
LogReg	accel spikes	20	window60	99.4	99.4	86.4	92.6	0.540	0.578
LogReg	accel spikes	40	whole	99.1	99.9	91.9	96.7	0.532	0.543
LogReg	accel spikes	40	window60	99.5	99.5	89.2	93.6	0.468	0.491
LogReg	fps cap	45	whole	99.1	97.0	91.9	86.3	0.532	0.549
LogReg	fps cap	45	window60	99.5	86.8	93.0	82.4	0.496	0.745
LogReg	fps cap	60	whole	99.1	98.1	91.9	88.4	0.532	0.535
LogReg	fps cap	60	window60	99.3	89.7	92.4	84.3	0.618	0.729
LogReg	fps cap	72	whole	99.1	98.6	91.9	90.2	0.532	0.534
LogReg	fps cap	72	window60	99.5	95.2	91.7	88.3	0.486	0.573
LogReg	fps restore	90	whole	99.1	99.1	91.9	91.9	0.532	0.532
LogReg	rot jitter	0.02	whole	99.1	99.2	91.9	92.1	0.532	0.533
LogReg	rot jitter	0.02	window60	99.5	99.5	91.5	91.6	0.499	0.500
LogReg	rot jitter	0.05	whole	99.1	99.3	91.9	92.5	0.532	0.533
LogReg	rot jitter	0.05	window60	96.4	96.8	92.2	92.5	0.673	0.668
LogReg	rot jitter	0.1	whole	99.1	99.2	91.9	92.7	0.532	0.535
LogReg	rot jitter	0.1	window60	95.7	96.9	91.1	91.3	0.629	0.605
LogReg	rot smooth	0.25	whole	99.1	99.1	91.9	91.7	0.532	0.532
LogReg	rot smooth	0.5	whole	99.1	99.1	91.9	91.7	0.532	0.532
LogReg	speed scale down	−0.5	whole	99.1	97.8	91.9	86.8	0.532	0.534
LogReg	speed scale down	−0.25	whole	99.1	98.7	91.9	89.2	0.532	0.526
LogReg	speed scale up	0.1	whole	99.1	99.2	91.9	92.9	0.532	0.536
LogReg	speed scale up	0.1	window60	99.4	99.4	87.8	88.7	0.513	0.518
LogReg	speed scale up	0.25	whole	99.1	99.4	91.9	93.6	0.532	0.536

Model	Scenario	Value	Window	Rec. clean	Rec. att.	FPR clean	FPR att.	Cost clean	Cost att.
LogReg	speed scale up	0.25	window60	99.4	99.4	90.8	91.9	0.520	0.527
LogReg	speed scale up	0.5	whole	99.1	99.7	91.9	94.9	0.532	0.537
LogReg	speed scale up	0.5	window60	99.5	99.5	86.0	92.5	0.452	0.486
RF	accel spikes	10	whole	98.7	99.0	81.7	84.5	0.485	0.494
RF	accel spikes	10	window60	99.3	99.3	75.0	79.1	0.444	0.467
RF	accel spikes	20	whole	98.7	99.7	81.7	96.2	0.485	0.543
RF	accel spikes	20	window60	97.6	99.1	68.6	96.2	0.466	0.605
RF	accel spikes	40	whole	98.7	99.8	81.7	97.4	0.485	0.549
RF	accel spikes	40	window60	99.5	100.0	81.1	99.8	0.427	0.511
RF	fps cap	45	whole	98.7	98.7	81.7	81.3	0.485	0.483
RF	fps cap	45	window60	99.4	99.4	84.9	84.9	0.456	0.456
RF	fps cap	60	whole	98.7	98.7	81.7	81.3	0.485	0.483
RF	fps cap	60	window60	99.2	99.2	84.7	84.7	0.569	0.569
RF	fps cap	72	whole	98.7	98.7	81.7	81.3	0.485	0.483
RF	fps cap	72	window60	99.4	99.4	85.6	85.6	0.457	0.457
RF	fps restore	90	whole	98.7	98.8	81.7	81.8	0.485	0.483
RF	rot jitter	0.02	whole	98.7	98.2	81.7	79.4	0.485	0.482
RF	rot jitter	0.02	window60	99.4	99.4	85.0	83.0	0.467	0.456
RF	rot jitter	0.05	whole	98.7	97.7	81.7	79.0	0.485	0.491
RF	rot jitter	0.05	window60	96.1	91.5	84.3	82.4	0.625	0.691
RF	rot jitter	0.1	whole	98.7	98.0	81.7	81.1	0.485	0.496
RF	rot jitter	0.1	window60	96.3	92.8	82.1	79.1	0.563	0.616
RF	rot smooth	0.25	whole	98.7	99.0	81.7	85.9	0.485	0.502
RF	rot smooth	0.5	whole	98.7	99.1	81.7	84.9	0.485	0.494
RF	speed scale down	−0.5	whole	98.7	90.6	81.7	61.8	0.485	0.553
RF	speed scale down	−0.25	whole	98.7	95.9	81.7	73.3	0.485	0.501
RF	speed scale up	0.1	whole	98.7	99.1	81.7	84.3	0.485	0.492
RF	speed scale up	0.1	window60	99.0	99.0	75.5	77.5	0.452	0.464
RF	speed scale up	0.25	whole	98.7	99.1	81.7	86.7	0.485	0.504
RF	speed scale up	0.25	window60	98.6	98.3	84.0	85.5	0.502	0.515
RF	speed scale up	0.5	whole	98.7	99.0	81.7	92.1	0.485	0.537
RF	speed scale up	0.5	window60	99.4	99.4	66.3	85.6	0.354	0.453

S5 Post-hoc analyses

All analyses in this section were specified after the primary results were known. They use the splits, inner folds, selection objective, checkpoint and threshold rules, cross-fitting recipe and final seeds of the primary protocol, and no primary result was changed. They differ in when they were specified. The exposure-time baselines (S5.1), the Holm adjustment of the comparisons with E-DDQN (S5.2), the comparison with always alarming (S5.3), the grouping sensitivity analyses (S5.4) and the cost curves (S5.5) were committed (`configs/posthoc.json`) before they were run. The paired comparisons of the exposure-time baselines with the primary models (main text, Section IV-B), the Holm adjustment over the nine models of the comparison with always alarming (S5.3) and the Q-value scale (S5.6) were added while the results were produced and written up, without prior commitment, as were two descriptive additions: the label-dynamics summary (S1.2) and the seed-ensemble block of the main results table. Sections S5.7–S5.10 were added after a pre-submission review, when all other results were known (key `review_round_2`); they use only the saved per-sample predictions, selected configurations and search logs. Section S5.11 was prompted by the same review but was specified, with its decision rules, and committed before any of its models was trained; like the exposure-time baselines of S5.1, it trains new models under the full protocol. Section S5.12, a descriptive calculation that trains no model, was added in the final pre-submission check. As in the main text (Section III-G), bootstrap p-values between 0.01 and 0.05 in the tables of this section are approximate, and a difference whose confidence interval includes zero is a point estimate only.

S5.1 Exposure-time baselines: selected configurations

Time-LR searched all six values of C; Time+game HGB used the HGB search space and budget (16 configurations).

Model	Outer fold	Feature set	Selected hyper-parameters	Inner-OOF cost
Time-LR	0	time	C = 0.001	0.471
Time-LR	1	time	C = 1	0.528
Time-LR	2	time	C = 0.1	0.549
Time-LR	3	time	C = 0.01	0.497
Time-LR	4	time	C = 0.1	0.550
Time+game HGB	0	time+game	lr 0.02, iter 200, leaves 8, min leaf 300, ℓ_2 1	0.480
Time+game HGB	1	time+game	lr 0.02, iter 200, leaves 8, min leaf 300, ℓ_2 0	0.501
Time+game HGB	2	time+game	lr 0.02, iter 100, leaves 4, min leaf 300, ℓ_2 10	0.524
Time+game HGB	3	time+game	lr 0.02, iter 200, leaves 4, min leaf 300, ℓ_2 1	0.522
Time+game HGB	4	time+game	lr 0.02, iter 200, leaves 8, min leaf 300, ℓ_2 0	0.506

S5.2 Holm adjustment of the paired comparisons

Group-bootstrap comparisons of E-DDQN with the six other primary models under each operating rule, with Holm's step-down adjustment over the six comparisons of each rule.

Rule	B	ΔCost (E-DDQN−B)	95 % CI	p_{group}	p_{Holm}
D1 (min cost)	E-CB	+0.175	[+0.016, +0.448]	0.014	0.070
D1 (min cost)	DQN	+0.190	[−0.017, +0.493]	0.079	0.105
D1 (min cost)	CS-MLP	+0.190	[+0.005, +0.485]	0.039	0.105
D1 (min cost)	LogReg	+0.194	[+0.011, +0.477]	0.026	0.104
D1 (min cost)	HGB	+0.183	[+0.011, +0.460]	0.035	0.105
D1 (min cost)	RF	+0.242	[+0.052, +0.528]	0.001	0.006
D0 ($\hat{p} \geq 1/6$)	E-CB	+0.214	[+0.025, +0.487]	0.018	0.054
D0 ($\hat{p} \geq 1/6$)	DQN	+0.242	[+0.039, +0.542]	0.012	0.048
D0 ($\hat{p} \geq 1/6$)	CS-MLP	+0.204	[−0.030, +0.507]	0.113	0.226
D0 ($\hat{p} \geq 1/6$)	LogReg	+0.087	[−0.293, +0.541]	0.694	0.694
D0 ($\hat{p} \geq 1/6$)	HGB	+0.328	[+0.078, +0.639]	0.003	0.015
D0 ($\hat{p} \geq 1/6$)	RF	+0.411	[+0.147, +0.748]	0.001	0.006
val. recall ≥0.95	E-CB	+0.220	[+0.017, +0.566]	0.021	0.080
val. recall ≥0.95	DQN	+0.236	[+0.013, +0.580]	0.028	0.080
val. recall ≥0.95	CS-MLP	+0.294	[+0.033, +0.652]	0.013	0.065
val. recall ≥0.95	LogReg	+0.307	[+0.027, +0.687]	0.020	0.080
val. recall ≥0.95	HGB	+0.314	[+0.030, +0.711]	0.023	0.080
val. recall ≥0.95	RF	+0.342	[+0.067, +0.719]	0.003	0.018

S5.3 Comparison with always alarming

Paired bootstrap of each model's D1 cost (seed ensemble) minus the cost of always alarming; Holm adjustment over the nine models (group units; the adjustment was added without prior commitment). With sessions instead of linkage groups as resampling units, which ignores possible dependence between linked sessions, the random forest's difference has $p = 0.002$ and Holm-adjusted $p = 0.018$ (approximate); all session-level values are in `results/tables/posthoc/vs_always_alarm.csv`.

Model	ΔCost (model−always)	CI (groups)	p	p_{Holm}	CI (sessions)	p
E-DDQN	+0.168	[−0.024, +0.466]	0.104	0.728	[−0.026, +0.438]	0.116
E-CB	−0.007	[−0.093, +0.103]	0.810	1.000	[−0.089, +0.105]	0.800
DQN	−0.022	[−0.104, +0.062]	0.560	1.000	[−0.099, +0.075]	0.598
CS-MLP	−0.022	[−0.046, −0.002]	0.037	0.296	[−0.049, +0.004]	0.109
LogReg	−0.026	[−0.061, +0.006]	0.136	0.816	[−0.058, +0.005]	0.113
HGB	−0.015	[−0.122, +0.121]	0.765	1.000	[−0.104, +0.092]	0.744
RF	−0.074	[−0.135, −0.019]	0.012	0.108	[−0.120, −0.025]	0.002
Time-LR	−0.010	[−0.073, +0.080]	0.737	1.000	[−0.080, +0.080]	0.782
Time+game HGB	−0.041	[−0.100, +0.051]	0.302	1.000	[−0.099, +0.048]	0.275

S5.4 Grouping sensitivity

Paired bootstrap resampling sessions instead of linkage groups (Holm over the six primary comparisons), and the range of the cost difference in a leave-one-group-out jackknife over the 21 groups.

B	ΔCost (E-DDQN−B)	CI (sessions)	p	p_{Holm}	jackknife range	sign flips
E-CB	+0.175	[+0.014, +0.423]	0.014	0.070	+0.068 to +0.201	0
DQN	+0.190	[+0.001, +0.454]	0.050	0.087	+0.084 to +0.248	0
CS-MLP	+0.190	[+0.006, +0.452]	0.037	0.087	+0.084 to +0.225	0
LogReg	+0.194	[+0.007, +0.454]	0.029	0.087	+0.094 to +0.225	0
HGB	+0.183	[+0.014, +0.440]	0.021	0.084	+0.076 to +0.215	0
RF	+0.242	[+0.055, +0.503]	0.001	0.006	+0.137 to +0.284	0
Time-LR	+0.178	[−0.015, +0.439]	0.089	–	+0.075 to +0.213	0
Time+game HGB	+0.209	[+0.016, +0.470]	0.030	–	+0.101 to +0.248	0

S5.5 Validation-chosen cost curves

Test cost of the seed ensemble divided by the cost of the better trivial policy, for each false-negative to false-positive ratio R (threshold chosen on inner-validation predictions at each R). At $R = 5$ the values reproduce the D1 seed-ensemble costs exactly (checked programmatically).

Model	1	1.5	2	3	4	5	7	10	15	20
E-DDQN	1.004	0.921	1.034	1.130	1.241	1.301	1.382	1.322	1.448	1.508
E-CB	0.828	0.819	0.900	0.920	0.954	0.988	1.068	0.992	1.007	1.013
DQN	0.921	0.796	0.865	0.845	0.929	0.961	1.013	1.068	1.208	1.304
CS-MLP	0.846	0.773	0.835	0.980	0.960	0.961	0.969	0.981	0.985	0.985
LogReg	0.852	0.868	0.915	0.941	0.946	0.953	0.966	0.983	0.989	1.010
HGB	0.897	0.770	0.808	0.884	0.945	0.973	1.001	0.980	0.991	0.998
RF	0.861	0.717	0.773	0.836	0.864	0.868	0.893	0.928	0.960	0.983
Time-LR	0.843	0.718	0.781	0.840	0.883	0.982	0.990	0.984	1.000	1.000
Time+game HGB	0.816	0.731	0.760	0.833	0.882	0.926	0.942	1.014	1.053	1.064

S5.6 Q-value scale

Largest absolute Q-value over the final runs of each outer fold relative to the return bound $r_{\max}/(1-\gamma)$ of the selected discount ($r_{\max} = 0.5$ for the asymmetric reward, 1 for the symmetric reward).

Model	Outer fold	γ	max $\lvert Q\rvert$	bound $r_{\max}/(1-\gamma)$	ratio
E-DDQN	0	0.99	41.45	50.00	0.83
E-DDQN	1	0.99	61.78	50.00	1.24
E-DDQN	2	0.99	35.08	50.00	0.70
E-DDQN	3	0.99	4.13	50.00	0.08
E-DDQN	4	0.9	11.56	5.00	2.31
E-CB	0	0	0.70	0.50	1.41
E-CB	1	0	0.68	0.50	1.37
E-CB	2	0	0.68	0.50	1.36
E-CB	3	0	0.89	0.50	1.79
E-CB	4	0	0.95	0.50	1.89
DQN	0	0	2.24	1.00	2.24
DQN	1	0.9	4.09	10.00	0.41
DQN	2	0.9	3.05	10.00	0.30
DQN	3	0.9	10.10	10.00	1.01
DQN	4	0	2.08	1.00	2.08

S5.7 Within-fold ROC-AUC

Pooled ROC-AUC combines the test scores of five fold-specific models and therefore mixes discrimination with differences in score scale between folds. Within-fold ROC-AUC is computed in each outer-test fold and averaged over folds (seed mean: also averaged over the five seeds).

Model	Pooled (seed mean)	Within-fold (seed mean)	Pooled (ensemble)	Within-fold (ensemble)
E-DDQN	0.538	0.539	0.541	0.552
E-CB	0.653	0.658	0.659	0.665
DQN	0.661	0.666	0.665	0.666
CS-MLP	0.644	0.666	0.654	0.672
LogReg	0.626	0.636	0.626	0.636
HGB	0.692	0.702	0.692	0.702
RF	0.688	0.715	0.688	0.715
Time-LR	0.704	0.709	0.704	0.709
Time+game HGB	0.676	0.699	0.676	0.699

S5.8 Exact group sign-flip tests

Two-sided sign-flip permutation test of the pooled D1 cost difference of the seed ensembles, with the 21 linkage groups as units and all 2^{21} sign patterns enumerated. Holm adjustment over the six primary comparisons with E-DDQN and over the nine comparisons with always alarming. The group-bootstrap p-values of Section S5.2 (rule D1) are shown for comparison.

Comparison (D1, seed ensembles)	ΔCost	p_{boot}	$p_{\text{boot,Holm}}$	$p_{\text{sign-flip}}$	$p_{\text{sign-flip,Holm}}$
E-DDQN vs E-CB	+0.175	0.014	0.070	0.048	0.241
E-DDQN vs DQN	+0.190	0.079	0.105	0.127	0.330
E-DDQN vs CS-MLP	+0.190	0.039	0.105	0.116	0.330
E-DDQN vs LogReg	+0.194	0.026	0.104	0.094	0.330
E-DDQN vs HGB	+0.183	0.035	0.105	0.083	0.330
E-DDQN vs RF	+0.242	0.001	0.006	0.014	0.083
E-DDQN vs Time-LR	+0.178	–	–	0.117	–
E-DDQN vs Time+game HGB	+0.209	–	–	0.067	–
Time-LR vs RF	+0.063	0.021		0.044	
Time+game HGB vs RF	+0.032	0.274		0.302	
E-DDQN vs always alarm	+0.168			0.250	1.000
E-CB vs always alarm	−0.007			0.922	1.000
DQN vs always alarm	−0.022			0.620	1.000
CS-MLP vs always alarm	−0.022			0.053	0.422
LogReg vs always alarm	−0.026			0.172	1.000
HGB vs always alarm	−0.015			0.769	1.000
RF vs always alarm	−0.074			0.022	0.194
Time-LR vs always alarm	−0.010			0.820	1.000
Time+game HGB vs always alarm	−0.041			0.358	1.000

S5.9 Selected feature sets and within-fold ROC-AUC

Exploratory. Feature set selected on validation data for each primary model and outer fold (T: telemetry; T+Q: telemetry and questionnaire indicators; E: exposure; E+Q: exposure and questionnaire indicators) and the within-fold ROC-AUC of that model on the fold's test groups (mean over seeds). Feature sets were selected, not assigned, so the association does not establish a cause.

Model	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4
E-DDQN	E+Q 0.541	E+Q 0.505	E+Q 0.723	E+Q 0.450	E+Q 0.476
E-CB	E 0.642	E 0.717	E 0.748	E 0.737	E+Q 0.446
DQN	E 0.706	E+Q 0.530	E 0.843	E 0.736	E+Q 0.515
CS-MLP	E+Q 0.562	E+Q 0.607	E 0.812	T 0.702	T 0.644
LogReg	E 0.623	T+Q 0.428	E 0.810	E 0.712	E+Q 0.610
HGB	E 0.631	E 0.740	E 0.791	T 0.761	E 0.585
RF	T 0.630	E 0.728	E 0.771	E 0.761	T+Q 0.683

S5.10 Search draw

Feature sets among each family's random-search candidates (T: telemetry; T+Q: telemetry and questionnaire indicators; E: exposure; E+Q: exposure and questionnaire indicators). The candidates were drawn once per family with a fixed seed and reused in every outer fold.

Model	Candidates	T	T+Q	E	E+Q
E-DDQN	16	4	2	2	8
E-CB	16	3	3	6	4
DQN	16	2	5	6	3
CS-MLP	16	4	5	5	2
LogReg	12	3	3	4	2
HGB	16	4	4	3	5
RF	16	4	3	4	5

For E-DDQN, the lowest pooled inner-validation 5:1 cost among candidates with (Q) and without the questionnaire indicators in each outer fold; the selection rule picks the lower value.

E-DDQN, outer fold	0	1	2	3	4
lowest validation cost, candidates with Q	0.424	0.452	0.484	0.490	0.483
lowest validation cost, candidates without Q	0.487	0.483	0.521	0.518	0.542

S5.11 Search-draw sensitivity (specified before it was run)

Prompted by the dependence of E-DDQN's result on its single random draw (S5.10), this analysis was specified in `configs/posthoc.json` (key `search_draw_sensitivity`), together with its decision rules, and committed before any of its models was trained. E-DDQN and the standard DQN were each tuned again under the full protocol—same splits, inner folds, search training seed, budget of 16 configurations, selection rule, checkpoint rule, cross-fitting, five final seeds and threshold rules—with two further random draws from the full search space (configuration seeds 9120 and 9130 for E-DDQN, 9122 and 9132 for DQN) and with one draw of 16 configurations restricted to the exposure feature set (9200 and 9202). Perturbation variants were not scored. Draw 1 is the primary analysis and is reproduced exactly. Every variant is reported; none replaces the primary results. The decision rules were: (R1) the statement that E-DDQN does not outperform the standard DQN is challenged if, in any pair, E-DDQN's seed-ensemble cost is lower and the 95 % group-bootstrap interval of the difference excludes zero; (R2) E-DDQN's result is called draw-dependent if its seed-mean cost or its within-fold ROC-AUC spans more than 0.05 over its three full-space draws; (R3) under the exposure-only restriction, a within-fold ROC-AUC of E-DDQN within 0.03 of DQN's supports the feature-set explanation of Section IV-E, one more than 0.05 below DQN's means that this explanation is insufficient, and anything in between is inconclusive.

Per variant: number of exposure-only candidates in the draw, number of outer folds whose selected configuration uses the questionnaire indicators (Q folds), 5:1 cost (rule D1; mean ± SD over seeds), seed-ensemble cost, pooled and within-fold ROC-AUC (seed means), and the discount factor selected in each outer fold.

Variant	E-only cand.	Q folds	$\text{Cost}_{5:1}$ (seed mean)	Ens. cost	ROC-AUC	Within-fold AUC	Selected γ
E-DDQN, draw 1 (primary)	2	5/5	0.663±0.048	0.726	0.538	0.539	0.99, 0.99, 0.99, 0.99, 0.9
DQN, draw 1 (primary)	6	2/5	0.584±0.040	0.536	0.661	0.666	0.0, 0.9, 0.9, 0.9, 0.0
E-DDQN, draw 2	8	0/5	0.549±0.014	0.527	0.711	0.714	0.99, 0.99, 0.9, 0.99, 0.0
DQN, draw 2	4	3/5	0.634±0.066	0.662	0.634	0.631	0.9, 0.0, 0.0, 0.5, 0.0
E-DDQN, draw 3	7	2/5	0.558±0.019	0.561	0.658	0.671	0.9, 0.9, 0.0, 0.0, 0.99
DQN, draw 3	3	4/5	0.580±0.034	0.558	0.628	0.629	0.9, 0.9, 0.9, 0.5, 0.9
E-DDQN, exposure only	16	0/5	0.565±0.021	0.520	0.676	0.687	0.9, 0.5, 0.9, 0.9, 0.5
DQN, exposure only	16	0/5	0.536±0.017	0.545	0.705	0.712	0.5, 0.0, 0.9, 0.5, 0.0

Paired comparisons of E-DDQN with the standard DQN within each draw (seed ensembles, rule D1; 2,000 group-bootstrap replicates; exact sign-flip test over all 2^{21} sign patterns; ROC-AUC difference of the seed-ensemble scores).

Draw	ΔCost (E-DDQN−DQN)	95 % CI	p_{boot}	$p_{\text{sign-flip}}$	ΔROC-AUC	95 % CI	p_{boot}
draw 1 (primary)	+0.190	[−0.017, +0.493]	0.079	0.127	−0.125	[−0.264, −0.018]	0.021
draw 2	−0.135	[−0.395, +0.024]	0.167	0.279	+0.088	[+0.021, +0.162]	0.009
draw 3	+0.004	[−0.052, +0.061]	0.953	0.907	+0.031	[−0.029, +0.115]	0.354
exposure only	−0.025	[−0.061, +0.025]	0.276	0.329	−0.025	[−0.066, +0.009]	0.156

Outcome under the pre-specified rules. R1 (headline challenged): no; no pair gave E-DDQN a significantly lower cost (differences −0.135 to +0.190; smallest group-bootstrap $p = 0.079$). R2 (draw-dependent): yes; over its three full-space draws E-DDQN's seed-mean cost spans 0.114 (0.549–0.663) and its within-fold ROC-AUC 0.175 (0.539–0.714), against 0.580–0.634 and 0.629–0.666 for the standard DQN. R3 (feature-set explanation): supported; with the exposure features only, the standard DQN's within-fold ROC-AUC exceeds E-DDQN's by 0.026. Beyond the rules, the ROC-AUC differences show that the ranking of the two agents reverses between draws: the standard DQN ranks seconds better in draw 1 ($p = 0.021$, approximate) and E-DDQN in draw 2 ($p = 0.009$). The variants were not compared with the other models by any test. The clean rerun of the final version reproduced every variant's per-sample predictions exactly (Section S7).

S5.12 Score scale of the untuned rule D0

Descriptive; added in the final pre-submission check; no model is trained (`scripts/14_score_scale.py`). The implied probability $\hat{p}$ (main text, Eqs. 5 and 7) is on the scale of $P(y = 1 \mid s)$ only for a model trained on the natural class mix with a squared or logistic loss. Class-balanced replay (E-DDQN, E-CB: a fraction ρ of non-events in every minibatch) and the class-balanced minibatches of the CS-MLP (50/50) train on a different class mix; the scores then estimate the posterior under that mix, whose odds are $k = \frac{1-\rho}{\rho} \cdot \frac{1-\pi}{\pi}$ times those of $P(y = 1 \mid s)$ ($k = \frac{1-\pi}{\pi}$ for 50/50 minibatches), where π is the event prevalence of the inner training split. The cut-off $\hat{p} \geq 1/6$ of rule D0 is then $P(y = 1 \mid s) \geq 1/(1 + 5k)$. This is exact when the scores are conditional means, as for E-CB ($\gamma = 0$, where the Huber loss is quadratic because the two rewards of each action differ by only 0.6) and for the CS-MLP; for E-DDQN ($\gamma = 0.9$–0.99) it is approximate, because some bootstrapped residuals exceed the threshold of the Huber loss. For the standard DQN, the Huber loss (threshold 1) with

the ±1 reward makes $\hat{p}$ a nonlinear function of $P(y = 1 \mid s)$; at $\gamma = 0$ the minimiser of the expected loss is $Q(s, 1) = -Q(s, 0) = (2p - 1)/\max(p, 1 - p)$, so $\hat{p} = 1/6$ at $P(y = 1 \mid s) = 0.25$ (no closed form for $\gamma > 0$, marked –). Logistic regression, gradient boosting, the random forest and the CS-MLP with uniform minibatches are trained on the natural class mix and keep the nominal cut-off 1/6 (up to calibration). The first table gives, for each selected configuration, ρ, the mean training prevalence π and k over the three inner splits, and the effective cut-off on $P(y = 1 \mid s)$ (mean and range over the three inner models; the main text quotes the range over all folds). Rule D0 and the greedy policies of the reward-ratio analysis depend directly on this scale. Rule D1 and the ranking metrics do not use the nominal cut-off, but they are not strictly invariant, because k differs between the three inner models whose validation scores are pooled for D1 and whose test scores are averaged, and between folds. The second table rescales every inner model of E-DDQN, E-CB and the CS-MLP to $P(y = 1 \mid s)$ with its own k (odds divided by k, extended monotonically to scores outside $(0, 1)$), re-derives D1 exactly as in the main analysis (whose saved decisions are first reproduced exactly) and compares the outcomes (means over the five seeds).

Model	Training class mix (selected)	Fold	ρ	π	k	Cut-off on $P(y = 1 \mid s)$: mean (range)
E-DDQN	class-balanced replay	0	0.400	0.457	1.786	0.101 (0.096–0.104)
E-DDQN	class-balanced replay	1	0.400	0.439	1.922	0.095 (0.084–0.101)
E-DDQN	class-balanced replay	2	0.400	0.437	1.951	0.094 (0.085–0.110)
E-DDQN	class-balanced replay	3	0.400	0.446	1.869	0.097 (0.091–0.105)
E-DDQN	class-balanced replay	4	0.500	0.436	1.305	0.135 (0.120–0.157)
E-CB	class-balanced replay	0	0.500	0.457	1.191	0.144 (0.137–0.149)
E-CB	class-balanced replay	1	0.500	0.439	1.282	0.136 (0.121–0.144)
E-CB	class-balanced replay	2	0.500	0.437	1.301	0.135 (0.123–0.156)
E-CB	class-balanced replay	3	0.600	0.446	0.831	0.195 (0.184–0.208)
E-CB	class-balanced replay	4	0.500	0.436	1.305	0.135 (0.120–0.157)
CS-MLP	class-balanced minibatches (50/50), class weight 5	0	–	0.457	1.191	0.144 (0.137–0.149)
CS-MLP	class-balanced minibatches (50/50), class weight 5	1	–	0.439	1.282	0.136 (0.121–0.144)
CS-MLP	natural class mix, class weight 5	2	–	0.437	–	0.167
CS-MLP	class-balanced minibatches (50/50), class weight 5	3	–	0.446	1.246	0.139 (0.131–0.149)
CS-MLP	natural class mix, class weight 5	4	–	0.436	–	0.167
DQN	±1 reward, Huber loss ($\gamma = 0.0$)	0	–	0.457	–	0.250
DQN	±1 reward, Huber loss ($\gamma = 0.9$)	1	–	0.439	–	–
DQN	±1 reward, Huber loss ($\gamma = 0.9$)	2	–	0.437	–	–
DQN	±1 reward, Huber loss ($\gamma = 0.9$)	3	–	0.446	–	–
DQN	±1 reward, Huber loss ($\gamma = 0.0$)	4	–	0.436	–	0.250

Model	D1 decisions changed (%)	$\text{Cost}_{5:1}$		Pooled ROC-AUC		Within-fold ROC-AUC	
		pipeline	rescaled	pipeline	rescaled	pipeline	rescaled
E-DDQN	0.78	0.663	0.656	0.538	0.535	0.539	0.536
E-CB	0.66	0.575	0.574	0.653	0.653	0.658	0.657
CS-MLP	0.08	0.557	0.556	0.644	0.643	0.666	0.666

S6 Reconstruction of the earlier protocol

An earlier, unpublished version of this study reported 96.88 % accuracy and 99.53 % recall for E-DDQN. Its 41-feature input files, preprocessing code and trained models were not available, so the following is a reconstruction of its protocol, not an exact reproduction. The earlier version's partition sizes (14,680 / 3,146 / 3,146; 20,972 in total; 3,670 training samples per grade) are reproduced exactly by oversampling every discomfort grade of the curated release to the size of grade 0 (5,243) and then splitting at row level.

Model	Evaluation protocol	Acc.	Recall	Spec.	MCC	ROC-AUC
HGB	Oversample to 5,243/grade, then 70/15/15 row split (earlier protocol)	97.6	99.8	91.0	0.935	0.995
HGB	Row-level random split, no oversampling	95.3	94.1	96.3	0.905	0.992
HGB	Grouped 5-fold split (session-linkage groups)	52.2	33.6	66.9	0.005	0.523
E-DDQN (earlier code)	Earlier script, earlier protocol (proxy)	95.2	99.2	83.1	0.870	–
E-DDQN (earlier code)	Earlier script, grouped split (proxy)	41.1	92.1	7.7	−0.004	–

Under this protocol 2,170 of the 2,359 positive test rows (92 %) have an exact duplicate in the training split, and all sessions occur in both splits.

Implementation pitfalls found in the earlier code

The following defects of the earlier E-DDQN implementation are listed because they are easy to make whenever a classifier is trained as a Q-learner. Items 1–3 are guarded against by the unit tests of the present code (Section S7); items 4–6 are avoided by design (the Double-DQN arg-max is computed in evaluation mode, the outer-test data are scored only once after training, and SHAP explains the decision margin ΔQ rather than

$Q(s, 1)$ on 400 test rows of each fold, with DeepExplainer and its additivity check tried first and expected gradients used when that check failed, as it did in every fold; main text, Sections III-E, III-G and III-H).

1. The next state was the next row of a shuffled chunk, with no terminal flags, so $\gamma = 0.99$ bootstrapped across unrelated samples.
2. The true label was stored as the action together with the reward of the random action actually taken (a form of teacher forcing).
3. The Polyak update ignored batch-normalization buffers, so the target network normalized with mean 0 and variance 1 throughout training.
4. The Double-DQN arg-max ran the online network in training mode.
5. The test set was scored during training (every 10 episodes), which invites test-set monitoring.
6. SHAP attributed $Q(s, 1)$, which contains $V(s)$, on 50 test rows with the additivity check disabled.

S7 Reproducibility

- Software: Python 3.13.16, PyTorch 2.14.1 (run on the CPU), scikit-learn 1.9.1, NumPy 2.5.3, pandas 3.0.5, SciPy 1.18.1, SHAP 0.53.0, statsmodels 0.15.0 (`requirements.txt`).
- One command regenerates everything from the raw XML files: `bash run_all.sh --fresh` (about 2.5 to 3.5 hours of wall time on four CPU cores, dominated by the reinforcement-learning hyper-parameter searches and ablation). A clean rerun of commit `2e49e93` from an empty directory, including the search-draw analysis, took about 2 hours 46 minutes of computing and reproduced every per-sample prediction, every reported number and the text of the paper and supplement of that commit exactly; it was interrupted once by a restart of the computing container and resumed from its saved per-job outputs, as recorded in `results/qc/clean_rerun.md`. Later commits changed the text and references, corrected the implementation of decision rule R3 of the search-draw analysis (its outputs are unchanged) and added the descriptive calculation of Section S5.12. With the final code, the steps after model training were rerun in the same clean copy; they reproduced every result file of this version exactly, apart from wall-clock timing values that the text does not use (`results/qc/final_version_verification.md`).
- The post-hoc analyses (Section S5) are recorded in `configs/posthoc.json`; Section S5 states which of them were committed before they were run and which were added later. They are produced by `scripts/11_posthoc.py` (`run` and `analyze`; seeds 801–802 for the post-hoc search, bootstrap seeds 5–7), except the score-scale calculation (Section S5.12), produced by `scripts/14_score_scale.py`, and the search-draw sensitivity analysis (Section S5.11, key `search_draw_sensitivity`), which is produced by `scripts/13_search_draw_sensitivity.py` (configuration seeds 9120, 9130, 9122, 9132, 9200 and 9202; bootstrap seeds 5 and 6).
- Seeds: split 20261012 (outer) and 20261013–20261017 (inner); hyper-parameter search 777 (+family index for configuration sampling); final training 1000–1004; perturbations 4242; bootstrap 11 (CIs) and 5 (paired tests).
- Every number in the manuscript and in this supplement is produced by `scripts/05_make_tables.py` (primary analysis) from `results/tables/*.csv`, which `scripts/04_aggregate.py` computes from the per-sample predictions in `results/predictions/`, or by `scripts/11_posthoc.py`, `scripts/13_search_draw_sensitivity.py` and `scripts/14_score_scale.py` (post-hoc analyses, Section S5). `scripts/08_consistency_check.py` recomputes confusion matrices, metrics and McNemar statistics with independent library implementations and checks them against the manuscript macros.
- Executable sanity tests of the corrected agent: `scripts/00_sanity_tests.py` (reward regret ratio, implied threshold, within-session transitions, terminal flags, reward/action consistency of replay, target-network buffer synchronisation, Q-values within 1.5 times the return bound on a toy problem, McNemar identity)); checks of the evaluation boundaries, the selection steps and the replay buffer on the real folds: `scripts/15_pipeline_verification.py` (Section S8).

S8 Verification of the evaluation boundaries and of the replay buffer

`scripts/15_pipeline_verification.py` checks the implementation on the real folds of the evaluation set, using the production code; it trains no reported model. 169 of 169 checks pass (`results/qc/pipeline_verification.md`).

S8.1 Splits and duplicated rows

For the outer split and for all 45 pairs of partitions of the 15 outer × inner splits (inner training, inner validation, outer test), no session, linkage group or exact questionnaire profile occurs in two partitions, each linkage group lies in exactly one outer fold, the inner folds partition exactly the development groups, and the saved split files equal the recomputed splits. No raw telemetry row, with or without elapsed time, and no derived feature vector of an evaluation partition is identical to a row of the corresponding training partition.

As a descriptive check (no pass/fail criterion), the median distance in the scaled telemetry feature space from an outer-test row to its nearest development row is 1.35–2.92 across outer folds, against 1.67–2.01 from a development row to the nearest row of another linkage group and 1.40–1.49 to the nearest other row of its own session. Test rows are thus about as close to the training data as unrelated groups are to each other; in two folds they are about as close as rows of the same session. Because rows of one session are only modestly closer to each other than rows of different groups in this feature space, this distance discriminates weakly: it gives no sign of near-duplicated rows across the boundary, but it cannot exclude that a test group contains the same person as a training group. An earlier version of this check required the test-to-training distance to exceed the distance between consecutive seconds; that criterion was ill-posed (a nearest neighbour among about 7,000 rows is naturally closer than one fixed neighbour), failed in four of five folds, and was replaced by the two reference distances above.

S8.2 Training-only fitting and selection

For every one of the 15 outer × inner splits, the learned preprocessing parameters (frame-rate imputation value, acceleration-spike threshold, scaler centres and scales) and the training matrix are unchanged when every telemetry value of the validation and test rows is replaced by noise. The preprocessing used during hyper-parameter search contains no outer-test rows, and none of the 3,390 search runs stored a test score. Every selected configuration is the best configuration by pooled inner-validation cost (tie-break PR-AUC), as recomputed from the search log; every D1 threshold, per seed and for the seed ensemble, is recomputed exactly from the saved inner-validation scores; and in all 300 final training runs of the neural models the checkpoint is the best inner-validation epoch, with no test metric in any training log.

S8.3 Replay transitions

E-DDQN, E-CB and the standard DQN were trained with each outer fold's selected configuration on the first inner split (15 runs, shortened to two epochs), and every transition written to the replay buffer was logged (240,646 transitions in total). In every run: the stored reward equals $R(a, y)$ for the stored action a and the label y of the stored state; every non-terminal next state is the next second of the same session; the terminal flag is set exactly at the last second of each session; and the buffer holds exactly the logged states, actions, rewards, next states and flags, so no transition joins two sessions and no action is overwritten after its reward is computed. The training rows of each session are contiguous and in temporal order. With full-information replay, half of the stored transitions carry the action that was not executed, each with its own reward (Section III-E of the paper); no stored transition pairs one action with the reward of another.

S8.4 What these checks do not show

They show that the evaluation is session- and linkage-group-disjoint and that every learned quantity comes from training or inner-validation data. They do not show that the evaluation is participant-independent: the release contains no participant identifiers, and a person who answered the questionnaire differently in two non-consecutive sessions would not be linked.

S9 Material moved from the main text of the shortened version

The shortened manuscript (`main_access_v2.tex`) moved the following secondary tables and details here. Their content and numbers are unchanged from the full-length version.

Table S4: Main terms and notation.

Term	Meaning
Session (episode)	One recorded play-through of one game, logged about once per second; one episode of the decision process.
Linkage group	Sessions that may come from the same person; the unit of every split.
Event second, y_t	$y_t = 1$ if the discomfort reported at second t is at least 1 (slight), otherwise $y_t = 0$.
State s_t	Causal feature vector at second t, computed from seconds $0, \ldots, t$ of the same session only.
Action a_t	1 = raise an alarm, 0 = stay silent.
Reward $R(a, y)$	$R(1,1) = +0.5$, $R(0,0) = +0.1$, $R(0,1) = -0.5$, $R(1,0) = -0.1$.
Regrets Δ_{FN}, Δ_{FP}	Reward lost by a missed event (1.0) and by a false alarm (0.2).
$Q(s,a)$, $\Delta Q(s)$	Action value; decision margin $Q(s,1) - Q(s,0)$.
Score $\hat{p}(s)$	A model's output mapped to the nominal scale of $P(y = 1 \mid s)$.
5:1 cost c	$(5\,\mathrm{FN} + \mathrm{FP})/N$.
OOF prediction	Out-of-fold prediction: the score of an inner-validation second from a model not trained on its group.
Operating rule	How the alarm threshold on $\hat{p}$ is chosen: D1 (minimum validation 5:1 cost; primary), D0 ($\hat{p} \geq 1/6$) or a validation recall constraint.

Table S5: Gaps in the evaluation of asymmetric-reward cybersickness detectors and how this study addresses them (references as in the main text).

Gap	Evidence	This study
Splits of 1-Hz session data without session separation	70/30 split without reported session separation on the same gameplay data (TruVR, Kundu et al.); leakage literature (Kapoor and Narayanan; Kaufman et al.; Roberts et al.)	Session-linkage groups, nested grouped evaluation, train-only preprocessing, protocol fixed before testing; effect of the split protocol reported
Relation of asymmetric rewards to cost-sensitive learning left implicit	DQN classifiers for imbalanced data (Lin et al.; Yang et al.)	Derivation from known results that the 5:1 reward equals the 5:1 cost-sensitive Bayes threshold, and of how error-triggered termination departs from it
RL detectors not compared with matched cost-sensitive or minimal baselines	–	Seven models with the same search procedure, number of configurations and decision rules; two exposure-time baselines; group-bootstrap and exact sign-flip inference with multiplicity adjustment
Robustness assessed with optimization-based input attacks only	Kundu et al.; Ahmed et al.	Bounded, physically motivated telemetry perturbations as a sanity check, with adaptive evasion and false-alarm worst cases

S9.1 Main terms and notation

S9.2 Research gaps

S9.3 Expected cost under three operating rules

S9.4 Matched operating points

Thresholds were chosen on validation predictions to reach the stated recall or specificity (descriptive, no test). The thresholds transfer worst for E-DDQN: at both recall targets shown, and also at 0.80 and 0.99 (released results), it attains lower test recall than every supervised model, and at the 0.95 target also a higher false-positive rate. Read from the ROC curves at fixed recall (descriptive), the false-positive rate at 95 % recall is 97.7 % for E-DDQN and 65.8 % for the random forest.

S9.5 Selection optimism

S9.6 Reward-ratio sensitivity

The regret ratio of E-DDQN's reward was varied from 1:1 to 10:1 with the selected configuration per fold fixed (ratio 5:1 reproduces the main E-DDQN run bit for bit). The nominal greedy cut-off is $1/(1 + R)$; under class-balanced replay it corresponds approximately to $1/(1 + kR)$ on $P(y = 1 \mid s)$, with $k = 1.07$–2.18. Recall of the greedy policy rises from 52.4 % (1:1) to 77.2 % (5:1) and 80.7 % (10:1), while specificity falls from 58.1 % to 28.0 % and 23.0 %. With validation-chosen thresholds all ratios end at similar costs (0.629–0.673; the largest difference is smaller than any ratio's seed SD; no formal test).

Table S6: Expected 5:1 cost (mean over seeds) under three operating rules chosen without test data.

Model	D1: min cost	D0: $\hat{p} \geq 1/6$	val. recall ≥ 0.95
E-DDQN	0.663	0.905	0.785
E-CB	0.575	0.663	0.581
DQN	0.584	0.680	0.573
CS-MLP	0.557	0.731	0.544
LogReg	0.532	0.808	0.518
HGB	0.543	0.567	0.511
RF	0.491	0.486	0.483
Always alarm	0.558	0.558	0.558

Table S7: Matched operating points: test recall (%), false-positive rate (%) and false-positive count of the seed ensemble at validation-chosen thresholds.

	val. recall ≥ 0.90			val. recall ≥ 0.95			val. specificity ≥ 0.80		
Model	Rec.	FPR	FP	Rec.	FPR	FP	Rec.	FPR	FP
E-DDQN	81.7	77.0	4,036	84.5	86.6	4,539	25.2	18.6	974
E-CB	88.6	68.0	3,565	92.4	78.1	4,093	31.2	17.4	910
DQN	88.0	64.2	3,366	91.7	72.7	3,810	32.2	14.2	743
CS-MLP	94.1	78.4	4,111	97.5	85.1	4,463	24.2	12.7	666
LogReg	92.8	73.5	3,856	97.2	81.7	4,285	37.7	22.5	1,181
HGB	95.4	72.2	3,785	97.5	81.7	4,282	24.3	11.3	594
RF	93.7	64.0	3,358	96.9	74.3	3,896	21.0	13.3	695

S9.7 Perturbation stress test: threshold-free ranking

Largest drops: acceleration-spike injection in E-CB and DQN, speed down-scaling in gradient boosting.

S9.8 SHAP attributions of the E-DDQN decision margin

We explain $\Delta Q(s) = Q(s,1) - Q(s,0)$ rather than a single Q-value, because $Q(s,1)$ also contains the state value $V(s)$, which is shared by both actions. One model per outer fold is retrained with the configuration, seed and inner split of that fold's first final model; 200 background samples are drawn from its training rows and attributions computed for 400 of its test samples. DeepExplainer failed its additivity check in every fold (it does not recognize the layer-normalization modules of the selected networks), so the expected-gradients estimator (GradientExplainer) was used; the method per fold is recorded with the released results. Features are named as in the released feature dictionary (Section S1.4).

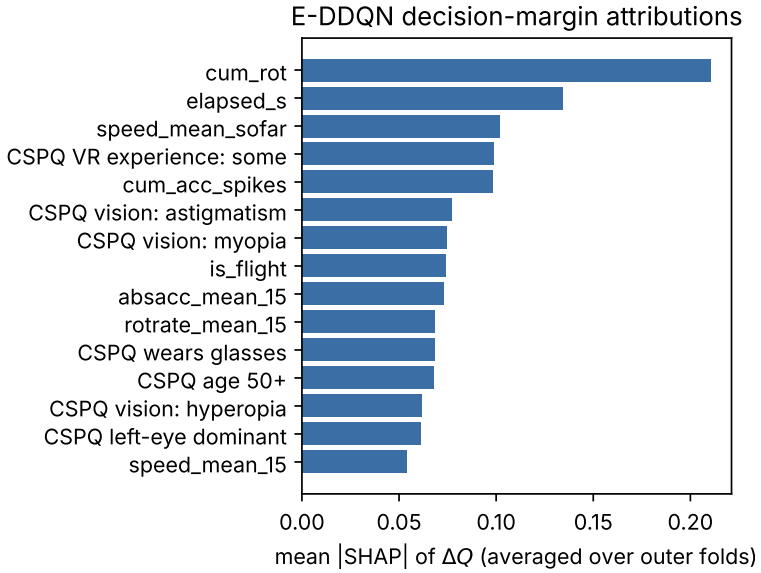


Figure S1: Mean absolute SHAP value of the E-DDQN decision margin ΔQ (primary search draw), averaged over the five outer folds.

Table S8: Pooled inner-validation 5:1 cost of the selected configuration (search seed, at its cost-minimizing threshold) versus its outer-test cost (seed ensemble, rule D1), averaged over the five outer folds (unweighted fold means, which differ from the pooled costs of the main results table).

Model	Inner-OOF cost at selection	Outer-test cost	Gap
E-DDQN	0.467	0.754	+0.288
E-CB	0.488	0.557	+0.070
DQN	0.478	0.544	+0.065
CS-MLP	0.501	0.534	+0.033
LogReg	0.518	0.531	+0.014
HGB	0.518	0.539	+0.021
RF	0.495	0.481	−0.015

Table S9: Reward-ratio sensitivity of E-DDQN: greedy (arg-max) policy and validation-tuned threshold (rule D1); mean (± SD) over five seeds.

		Greedy policy ($\arg\max_a Q$)			Validation-tuned threshold (D1)			
FN:FP regret	implied cut-off	Recall	Spec.	$\text{Cost}_{5:1}$	Recall	Spec.	$\text{Cost}_{5:1}$	ROC-AUC
1:1	1/2	52.4±5.6	58.1±4.8	1.285	94.9	7.4	0.629±0.056	0.558
2:1	1/3	69.2±3.3	43.1±1.6	0.999	94.6	6.4	0.642±0.074	0.563
5:1	1/6	77.2±3.8	28.0±2.0	0.905	93.4	7.4	0.663±0.048	0.538
10:1	1/11	80.7±3.2	23.0±3.3	0.855	92.9	7.4	0.673±0.095	0.552

S9.9 Implementation pitfalls of the earlier code

Three errors are easy to make when a classifier is trained as a Q-learner, and all three were present in an earlier implementation of this study: (i) transitions built from consecutive rows of shuffled data without terminal flags, so that the next state is an unrelated second, usually of another session; (ii) the true label stored as the action while the reward of a different, exploratory action is computed (a form of teacher forcing that trains the correct action toward the penalty of the wrong one); (iii) only the parameters averaged into the target network, so that its batch-normalization statistics never leave their initial values. Unit tests guard against each, and Section S8 verifies the replay buffer on the real folds.

Table S10: Pooled outer-test ROC-AUC of the seed-ensemble score on clean telemetry and under whole-session perturbations (last column: most negative and most positive change over all 17 whole-session variants, including the evasion-only candidates).

Model	Clean	fps≤45	jitter 0.10	speed ×1.5	spikes 40	speed ×0.5	smooth 0.25	min / max Δ
E-DDQN	0.541	0.541	0.593	0.561	0.559	0.532	0.539	−0.008 / +0.053
E-CB	0.659	0.659	0.682	0.648	0.575	0.648	0.658	−0.089 / +0.052
DQN	0.665	0.665	0.677	0.680	0.605	0.632	0.676	−0.064 / +0.035
CS-MLP	0.654	0.649	0.654	0.630	0.652	0.629	0.646	−0.025 / +0.005
LogReg	0.626	0.677	0.639	0.620	0.629	0.608	0.639	−0.017 / +0.051
HGB	0.692	0.694	0.709	0.650	0.698	0.647	0.712	−0.068 / +0.019
RF	0.688	0.689	0.708	0.667	0.683	0.723	0.725	−0.021 / +0.037